5. Expand the following squares, using suitable identities.

(i) $(2x+5)^2$ (ii) $(b-7)^2$ (iii) $(mn+3p)^2$ (iv) $(xyz-1)^2$

6. Using the identity $(a+b)(a-b)=a^2-b^2$, find the following product.

(i) $(p+2)(p-2)$

(ii) $(1+3b)(3b-1)$

(iii) $(4-mn)(mn+4)$

(iv) $(6x+7y)(6x-7y)$

7. Evaluate the following, using suitable identity.

(i) 51^2 (ii) 103^2 (iii) 998^2 (iv) 47^2

(v) 297×303 (vi) 990×1010 (vii) 51×52

8. Simplify: $(a+b)^2-4ab$

9. Show that $(m-n)^2+(m+n)^2=2(m^2+n^2)$

10. If $a+b=10$ and $ab=18$, find the value of a^2+b^2

11. Factorise the following algebraic expressions by using the identity $a^2-b^2=(a+b)(a-b)$.

(i) z^2-16 (ii) $9-4y^2$ (iii) $25a^2-49b^2$ (iv) x^4-y^4

12. Factorise the following using suitable identity.

(i) $x^2-8x+16$

(ii) $y^2+20y+100$

(iii) $36m^2+60m+25$

(iv) $64x^2-112xy+49y^2$

(v) $a^2+6ab+9b^2-c^2$

Objective Type Questions

13. If $a+b=5$ and $a^2+b^2=13$, then $ab=$?

(i) 12 (ii) 6 (iii) 5 (iv) 13

14. $(5+20)(-20-5)=$?

(i) −425 (ii) 375 (iii) −625 (iv) 0

15. The factors of x^2-6x+9 are

(i) $(x-3)(x-3)$ (ii) $(x-3)(x+3)$ (iii) $(x+3)(x+3)$ (iv) $(x-6)(x+9)$

16. The common factors of the algebraic expressions ax^2y, bxy^2 and $cxyz$ is

(i) x^2y (ii) xy^2 (iii) xyz (iv) xy

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